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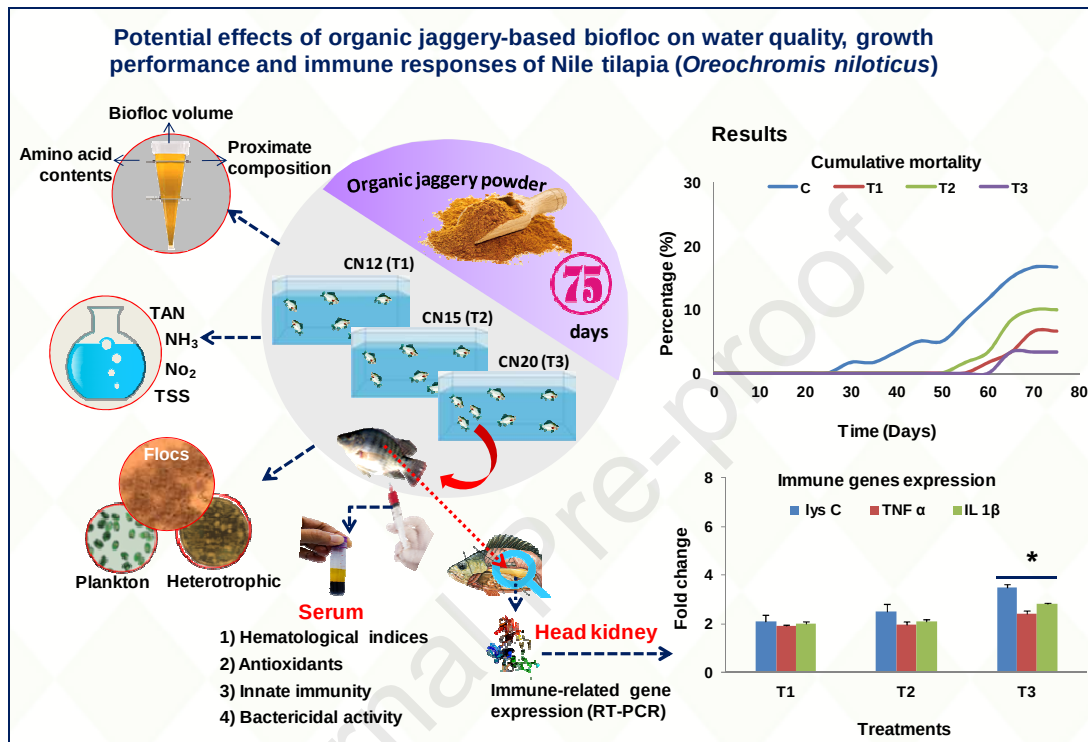
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Potential influence of jaggery-based biofloc technology at different C:N ratios on water quality, growth performance, innate immunity, immune-related genes expression profiles, and disease resistance against *Aeromonas hydrophila* in Nile tilapia (*Oreochromis niloticus*)

Sivaramasamy Elayaraja^{1,2,4*}, Mahmoud Mabrok³, Abdelazeem Algammal⁵, Elayaraja Sabitha², Mayavan Veeramuthu Rajeswari⁶, Kamil Zágorský⁴, Zhangying Ye¹, Songming Zhu^{1*}, Channarong Rodkhum^{2*}

¹College of Biosystems Engineering and Food Sciences, Zhejiang University, Hangzhou, China.

²Fish Infectious Diseases Research Unit (FID RU), Department of Veterinary Microbiology, Faculty of Veterinary Science, Chulalongkorn University, Bangkok, Thailand.

³Department of Fish Diseases and Management, Faculty of Veterinary Medicine, Suez Canal University, Egypt.

⁴Technical University of Liberec, Liberec, Czech Republic.

⁵Department of Bacteriology, Mycology and Immunology, Faculty of Veterinary Medicine, Suez Canal University, Egypt.

⁶Center for Ocean Research, Institute of Science and Technology, Sathyabama University, Chennai, India.

Correspondences*

Songming Zhu; E-mail: zhusm@edu.cn

Channarong Rodkhum; E-mails: channarong_r@yahoo.com

Sivaramasamy Elayaraja; E-mail: elayabact@yahoo.com

Data Availability

The authors declare that they do not have any shared data available

Conflicts of Interest

The authors declare that there are no conflicts of interest.

Abstract

Biofloc technology is increasingly becoming the most promising aquaculture tool especially in places where water is scarce and the land is very expensive. The dynamics of water quality, as well as plankton and microbial abundance, are collectively necessary for successful fish farming. The prospective use of jaggery as a potential carbon source and its influence on water quality, growth performance, innate immunity, serum bactericidal capacity, and disease resistance to *Aeromonas hydrophila* was investigated in *Oreochromis niloticus*. A completely randomized design was used in triplicates, where the control group was reared in a water system with no carbon source, while T1, T2, and T3 groups were raised in biofloc systems at C:N ratios of C:N12, C:N15, and C:N20, respectively. Water specimens were collected daily and fortnightly, while blood, serum, and head kidneys were collected at 75 days of experimental period for further analysis. TAN, nitrite, and ammonia values were considerably reduced, while the TSS values elevated significantly in all treated groups compared to the control. Jaggery-based biofloc system (JB-BFT) has a pronounced effect on hematological and growth performance parameters rather than control. Similarly, serum antioxidants, lysozyme, protease, antiprotease and bactericidal capacity were significantly increased ($p < 0.05$) in the treated groups in a dose-dependent manner. LYZ, TNF- α , and IL-1 β genes were upregulated in proportion to C:N ratios with the highest fold in C:N20. Furthermore, fish treated with JB-BFT presented lower cumulative mortalities and better relative levels of production (RLP) after experimental challenge with *A. hydrophila* compared to control. In conclusion, JB-BFT has a robust influence on Nile tilapia (*O. niloticus*) innate immunity through favorable innovation of various immune-cells and enzymes as well as upregulating the expression levels of immune-related genes. This study offers jaggery as a new carbon source with unique properties that satisfy all considerations of biofloc technology in an eco-friendly manner.

Keywords: biofloc, experimental challenge, innate immunity, jaggery powder, Nile tilapia.

1. Introduction

Tilapia is one of the most popular fish species globally due to its outstanding traits such as relative palatability, fast-growing, ease of cultivation, and resistance to the adverse environmental conditions [1]. Over the past few decades, the unprecedented world population has explained various plausible scenarios for aquaculture expansion and fish intensification, which may dissipate the limited water resources and increase environmental deterioration [2]. A provisional nutrient balance for intensive fish farming could produce enormous types of organic-pollutants which induced severe toxicity and gradual environmental degradation [3]. Despite the inspiring advances in science and technology, many fish farming sectors are still implementing outdated management protocols, including a high rate of water exchange and overuse of chemical disinfectants, which have many limitations for aquaculture sustainability and economic efficiency [4].

Recently, there is a consensus on finding efficient alternative rearing systems to increase fish production using limited, high-quality water resources [5]. Recirculating aquaculture systems (RAS) are among the most widely used to reduce major pollutants from culture water without any environmental degradation. However, the adoption of this technology among the agricultural community, especially in small scale sectors is lamentable due to high costs of operation and maintenance [6]. Therefore, several approaches have been performed to find out a sustainable cost-effective technology with a high level of nutrients for wide scales adoption. Biofloc technology (BFT) is increasingly becoming a promising eco-friendly aquaculture tool in which the nutrients could be recycled and constantly reused [7]. The technology works on the basic principle of bio-flocculation, where the retention of organic nitrogenous wastes converting into a proteinaceous compound, which could be utilized by the fish as natural food source [8]. Based on that, the widespread adoption of BFT is economically feasible, as the system can operate at no cost and does not require an external water treatment system [9].

BFT has been extensively used for culturing various aquatic species including fish and shrimp where the water exchange is limited and the microbial immobilization of ammonium is required, all of which are essential practices for sustainable aquaculture production [10]. BFT not only regulates the feed efficiency and nutrient utilization of cultured aquatic species [11] but also provides extra nutrients and exogenous enzymes to improve digestion [12]. The bacterial species included in biofloc systems play a noticeable immunological effect on the immune response of

fish due to the bacterial cell wall components such as β -1,3,-glucans, LPS and peptidoglycan, which independently or synergistically activate the innate immunity of fish [13].

Carbon amendment is a key factor for the ammonia pathway in BFT systems [14]. At high C:N ratios, heterotrophic bacteria are the most predominant species, while at low ratios a mixed bacterial community of heterotrophic and autotrophic bacteria dominates [15]. Unlike autotrophs, the heterotrophic bacteria play a chief role in stabilizing the system and maintaining water quality through a consecutive process of ammonia immobilization as well as generating *in situ* microbial protein [16]. The C/N ratio was manipulated by the frequent addition of carbohydrate or by elevating the carbon level in fish feed [17]. Recently, the application of biofloc technology as a basic system for fish and shrimp farming has received considerable attention and is generally beneficial [10, 18]. However, the major hurdles in the application of various sugars in BFT are the high investment costs and strong competition between the carbon sources. Most of the sugars used in BFT have not been greatly exploited especially in developing countries due to their high economic impact [19]. Hence, it was necessary to explore a new sugar with unique features such as ease of availability, reasonable price, and effective bacterial assimilation for sustainable and profitable aquaculture.

Jaggery, a sugarcane product, contain several essential components, including minerals (*viz* Calcium-40-100 mg, Magnesium-70-90 mg, Potassium-1056 mg, Phosphorus-20-90 mg, Sodium-19-30 mg, Iron-10-13 mg, Manganese-0.2-0.5 mg, Zinc-0.2- 0.4 mg, Copper-0.1-0.9 mg, and Chloride-5.3 mg per 100 g of jaggery), vitamins (*viz* Vitamin A-3.8 mg, Vitamin B1-0.01 mg, Vitamin B2-0.06 mg, Vitamin B5-0.01 mg, Vitamin B6-0.01 mg, Vitamin C-7.00 mg, Vitamin D2-6.50 mg, Vitamin E-111.30 mg, Vitamin PP-7.00 mg), protein (0.28%), carbohydrates (Sucrose-72-76 g, Fructose-1.5-7 g, Glucose-1.5-7 g), water 1.5-7 g, and calories 312 mg [20], which can be used as sweeteners and/or dietary supplements for animal feed mixtures [21]. Jaggery was recently introduced as a new carbon source to shrimp biofloc technology and showed a significant increase in microbial growth, biofloc nutritional values, and gave promising results in shrimp growth performance and survival rate compared to other carbon sources, including molasses and sugar [22]. Singh et al. [23] reported that jaggery improves the growth of heterotrophic bacteria, which are the primary component of biofloc, as it provides the basic energy needed for microbial cells growth. Moreover, the sugar is naturally loaded with antioxidants and vital minerals such as zinc and selenium, which prevent free radical damage and

thus strengthen immunity and boost resistance against infections [24]. Therefore, this study seeks to address the potential effects of jaggery as a new carbon source on water quality, growth performance, innate immunity, and disease resistance to bacterial insults in Nile tilapia reared in zero water exchange system.

2. Materials and Methods

2.1. Animal sampling and ethics

All experiments were performed by expert scientists following IACUC guidelines on animal welfare and use for scientific purposes and were approved by the Institutional Animal Care and Use Committee (IACUC) of Zhejiang University, Zhejiang, China.

2.2. Biofloc preparation and maturation

A total of 150 mono-sex Nile tilapia weighing 30 ± 2.1 g were collected from a private fish farm in Guangdong, China and left acclimated for 35 days in a big round-shaped fiber tank of 1.5 m^3 capacity and 4.5 kg stocking density. The tank was filled with sand-filtered pond water and covered with a dark cloth to block the sunlight. Dissolved oxygen (5 ± 1 mg/L) was supplied by air-stones connected to 2 air pumps (185 W, ACO-012, Sunsun, Zhejiang, China); whereas the pH value was adjusted at 7.5 ± 0.5 . The fish were fed 3 times daily (09:00, 14:00 and 20:00 h) at a rate of 5% of their body weight on a commercial pellet of 30% crude protein (Tongwei, Chengdu, China). For biofloc preparation, the tank was supplied with jaggery powder (JP) (Xi'an Henrikang Biotech.co.Ltd), purchased from a commercial seller, Zhejiang province, China. The JP was added daily to induce a heterotrophic medium and promote biofloc maturation. Bioflocs volume (BFV) was measured weekly on-site using Imhoff cones [25].

2.3. Experimental setup

A total of 240 newly purchased, healthy mono-sex tilapia were randomly distributed into four identical groups after 14 days of acclimation period. Each group contributes 3 small fiber tanks of 100 L capacity and 20 fish stocking density. Prior to the experiment, the fish were starved for 24 h, weighed to manage the initial biomass values. The fish of the first group was reared in dechlorinated, sand-filtered, and UV-saturated freshwater with no further addition of carbon sources (control group, C). Fish of other groups were supplied with initially prepared biofloc water, where the C:N ratios were maintained at C:N12, C:N15, and C:N20 (T1, T2, and T3,

respectively) using an JP as a source of carbohydrate (see below). Different C: N ratios were selected based on our preliminary study, since ratios below C: N12 and above C: N20 showed disappointing results in terms of growth performance and survival rate (data not shown). The control fish were fed twice daily (9:00 and 14:00 h.) at a rate of 5-3% of their body weight, while the treated ones were fed to apparent satiety, and the experiment lasted for 75 days. Representative samples of fish were weighed fortnightly, and the feeding amount was adjusted accordingly. All tanks were strongly aerated using the same devices mentioned above. No water changes were performed in the biofloc groups, except to correct evaporation losses, while about 40% of the water was changed fortnightly in the control group.

2.4. Carbohydrate calculation

To maintain the different C:N ratios (C:N12, C:N15, and C:N20), the amount of carbon consumed per treatment was calculated based on the protein content (%) of the commercial feed used and the total nitrogen excreted in the environment according to Lima et al. [26] with some modifications.

The following equations were used:

Q_{Feed} (quantity of feed, g = Fish initial weight (g/ biomass) x feeding rate (%))

N_{feed} (amount of nitrogen produced, g) = $Q_{\text{Feed}} \times \text{crude protein\%} \times 6.25^{-1}$, as each g of feed having 16% of nitrogen

$N_{\text{Excretion}}$ (amount of nitrogen excreted, g) = $N_{\text{feed}} \times 70\%$, as only 70% of produced nitrogen will be excreted, while the rest was digested and transformed into muscle.

Amount of carbohydrate (g) = $N_{\text{Excretion}} \times [(C:N)] \times \%C$ (%C: the percentage of carbon in the utilized carbohydrate, 41.5% in jaggery).

The amount of jaggery was added regularly to maintain C:N ratio during the experiment, taken into account the amount of water added to correct evaporation losses as well as the volume of matured bioflocs.

2.5. Water quality assessment

Throughout the 75 days experimental period, dissolved oxygen (DO), pH, and water temperature were estimated daily by the aid of a portable digital probe meter (FG4-FK Mettler Toledo, Zurich, Switzerland). For other water quality parameters, 100 mL of water samples were

aseptically gathered from each tank fortnightly and used for further analysis [27]. Subsequently, half of the water samples were inspected for nitrate-nitrogen ($\text{NO}_3\text{-N}$) nitrite-nitrogen ($\text{NO}_2\text{-N}$), and total ammonia-nitrogen (TAN) using a spectrophotometer [28], while the residual samples were filtered under vacuum through pre-weighed and dried Whatman GF/C filter paper to measure total suspended solids (TSS) [29]. Bioflocs volume (BFV) was measured on-site by the aid of Imhoff cones (after 30 min sedimentation of 1000 mL of water specimens) [25]. Total plankton was measured using a plankton net (mesh size 45 μm) and was expressed as cells/mL. Total heterotrophic bacteria (THB) were counted on TSA media (Oxoid, Hampshire, UK) using the spread plate technique. The inoculated plates were incubated for 48 h at 37 °C and the THB was expressed as colony-forming units per milliliter (CFUs/ mL). All the measurements were performed fortnightly in triplicates from all experimental groups.

2.6. Proximate composition of biofloc

Bioflocs were collected from each tank using 100- μm mesh size and dried at 60 °C for further biochemical analysis at the end of the experiment [30]. A known quantity of samples was dried at 105 °C until a constant weight achieved. Weight differences between primary and dried products represent moisture content in a percentage. For ash contents, the obtained dried samples were burnt at 550 °C in a muffle furnace for 3 h and were weighed after being cooled at room temperature. The crude protein and crude lipid contents were estimated by the Kjeldahl technique and fat extraction system, respectively, while the crude fiber content was measured using the automatic fiber analysis system. All amino acid constituents were investigated using high-performance liquid chromatography (HPLC) following the official laboratory protocol of Kebiao, Qingdao, Shandong, China.

2.7. Growth performance measurements

All growth markers (feed intake, body weight gain, feed conversion ratio; FCR, and specific growth rate; SGR) were calculated after 75 days of the experimental period following the instructions of Sivaramasamy et al. [31] and Ismail et al. [32] using the following formulas:

WeightGain (WG) = final weight (g/biomass) – initial weight (g/ biomass).

Feed conversion ratio (FCR) = feed intake (g)/Weight gain (g).

SGR = (In final weight (g) – In initial weight (g) $\times$ 100)/T (in days).

2.8. Sample collection

Prior sampling, 4 fish /tank were gathered and anesthetized using tricaine methanesulfonate (MS-222) at 120 mg/L. Subsequently, the blood was drawn from the caudal vein and then collected in heparin-coated vials. Aliquot of fresh blood was utilized to evaluate the blood profile [33], while the remaining blood was left to clot, centrifuged at $3000 \times g$ for 15 min to collect the serum. The designated serum was carefully removed using a sterile pipette and kept frozen at -20°C for later analysis. Simultaneously the head kidneys were collected aseptically for RNA extraction and gene expression analysis. Briefly, the fish were sacrificed under cold temperature using ice and the intestines and viscera were removed to expose the kidneys. Fresh head kidney samples (50-100 mg) were then harvested in sterile 1.5 ml Eppendorf tubes for further analysis.

2.9. Hematological profiles

The total red blood cells (RBCs) and white blood cells (WBCs) counts were performed using a Neubauer hemocytometer [34]. The RBCs and WBCs values were presented as 10^6 cells/ μl and 10^4 cells/ μl , respectively. Haemoglobin was measured using a commercial kit (Hemoglobin A1C TEST kits, China), while the Haematocrit values were measured using heparinized capillary tubes and expressed as a percentage fraction. All measurements were performed at the end of the experiment.

2.10. Serum antioxidants and innate immune parameters

After serum collection, part of the serum was utilized to determine the antioxidant levels of superoxide dismutase (SOD), catalase (CAT), malondialdehyde (MDA), and glutathione (GSH) using commercial kits (Nanjing Jiancheng Institute, Nanjing, China) and was estimated according to the manufacturer's instructions.

The lysozyme assay was performed using 20 μL of serum samples according to the turbidimetric technique previously reported by Guardiola et al. [33] with slight modifications. The measurement was determined at an optical density (OD) of 450 nm (at 35°C after 20 min) in a microplate reader (ThermoFisher Scientific, Darmstadt, Germany). Serial dilutions of hen egg-white lysozyme (Oxoid, Hampshire, UK) were utilized to perform a standard curve, where the lysozyme activity for each specimen was estimated and expressed as $\mu\text{g/ml}$.

Protease activity was conducted using 10 µL of serum samples based on the azocasein hydrolysis assay [35]. The protease activity was measured as a percentage using the standard formula:

$$\text{Protease activity (\%)} = ((\text{samples Abs} - \text{C:N Abs}) / \text{CP Abs}) \times 100.$$

Where, the C:N, negative control, phosphate buffer saline (PBS, 0% activity), while CP, positive control, trypsin (5 mg/mL; Sigma) (100% activity).

The total anti-protease activity was carried out using 10 µL of serum samples following the protocol of Ellis [36]. PBS instead of serum and trypsin was served as a blank whereas in place of serum alone was used as a reference sample. The percentage inhibition of trypsin activity compared to reference sample was then calculated.

Concerning the serum bactericidal activity, it was conducted using a 3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium (MTT) according to Ismail et al. [32] with minor modifications. The test is based upon the reduction of a yellow-colored MTT by living bacteria into blue formazan and the intensity of the formed color is directly proportional to the total viable bacterial cells. *Aeromonas hydrophila* and *Streptococcus agalactiae* were used for the current investigation. The isolates were kindly supplied by the Fish Infectious Research Unit, Department of Veterinary Microbiology, Faculty of Veterinary Science, Chulalongkorn University, Bangkok, Thailand, and were routinely cultured in tryptic soy broth (Oxoid, Hampshire, UK) for 24 h at 28 °C. The bacteria were identified using different sets of biochemical reactions, MALDI-TOF, and DNA-based molecular tools, where the species identity was confirmed by 16S rDNA sequencing and sequence alignment using NCBI basic Local alignment search tool (data are not presented). The spectrum was obtained by an automated 96-channel microtiter plate spectrophotometer interfaced to a computer at 560 nm wavelength. The bactericidal activity was estimated and referred to the proportion of viable bacteria in relation to the total number of bacteria in the positive control (serum-free templates). The analysis was conducted in triplicates at the end of the experiment.

2.11. Gene expression analysis

After serum collection, head kidneys were collected aseptically from dissected fish for further analysis. Total RNA was isolated using TRIzol™ Reagent (ThermoFisher Scientific, Darmstadt, Germany) following the manufacturer's instructions. Briefly, the collected head kidneys (50–100 mg) were gently meshed with 1 mL TRIzol using sterile plastic bars and mixed properly by

vortex. Subsequently, the specimens were mixed with RNase-free DNase (TAKARA, Japan) to remove the DNA remnant, and the first-strand cDNA (containing 1 µg RNA) was synthesized using First-Strand cDNA Synthesis Kit (ThermoFisher Scientific, Darmstadt, Germany) following the manufacturer's specifications. RNA purity and concentration were estimated by a NanoDrop 1000 spectrophotometer detection (ThermoFisher Scientific, Darmstadt, Germany) (A260/A280) and electrophoresis on a denaturing agarose gel 1%, where the ratios of 1.8-2 (260 nm/280 nm) were usually accepted as good candidates for RNA integrity.

Three common immune-related genes (LYZ, TNF- α , and IL-1 β) were selected and the transcriptional level of the tested gene was normalized to the expression of the β -actin reference gene, as the gene is constitutively expressed separately of treatments [37]. The list of primers with their Genbank IDs was illustrated in Table 1. The primer efficiency was estimated from the slope of the regression line after blotting the cycle thresholds (Ct) against the cDNA relative concentration [38]. Reaction volumes of 20 µL were amplified in a Mastercycler RealPlex4 PCR system (Eppendorf) utilizing 10 µL of IQ SYBR Green Supermix (Bio-Rad), 1 µL of diluted cDNA template (1:5 dilution), and 0.4 µL of a specific primer (10 µM). The recycling conditions including; 94 °C for 5 min followed by 35 cycles of 94 °C for 30 sec, 50-60 °C for 30-45 sec, and 72 °C for 30-45 sec.

Data are shown as fold change units, estimated by the following formula:

Fold change units = normalized values of the treated sample/ the normalized values of the control. All reactions were performed as technical triplicates.

2.12. Experimental challenge (Pathogenecity)

It was conducted against *A. hydrophila* as bacteria are one of the most serious pathogens affecting Nile tilapia and showed higher resistance to serum enzyme activity as shown in Table 7. At the end of the experimental period (75 days), 120 of the remaining live fish (30 from each treatment) were distributed into three groups, each contributes three identical subgroups ($n=10$). Only one subgroup was injected intraperitoneally (IP) with 0.2 ml of sterile normal saline, these subgroups were served as a negative control subgroups (C -ve). The remaining subgroups (I and II), which considered as duplicate infected subgroups for the C, T1, T2, and T3 experimental groups, were injected IP with 0.2 ml of virulent *A. hydrophila* at a concentration of 3×10^8 CFU/mL. For inoculum preparation, bacteria were routinely cultured as mentioned above (section

2.10. material and methods) and subsequently adjusted to the final concentration using a 0.5 McFarland standard and by means of the Helber counting chamber. All fish were monitored for 14 days post-challenge for clinical signs and cumulative mortalities. Moribund and/or freshly dead fish were aseptically collected and examined to set the cause of death. Mortalities were considered only when the injected strain was retrieved from infected fish (Koch's postulates). The recovered bacteria were identified biochemically and molecularly and the results showed that they were identical to the well-identified injected bacteria. The relative level of protection (RLP) was calculated following the Equation described by Ruangpan et al. [39]:

$$\text{RLP (\%)} = (1 - (\text{mortality (\%)} \text{ of biofloc treated group} / \text{mortality (\%)} \text{ of untreated group})) \times 100.$$

2.13. Statistical analysis

All data are provided as mean $\pm$ standard error of the mean (*SEM*). All values were performed as technical triplicates ($n= 3$). Data were analyzed by one-way ANOVA, followed by Tukey's post hoc test through paired-comparisons to determine significant variations between the experimental groups. Data normality and homogeneity were analyzed using the Kolmogorov-Smirnov test of variance and Levene's test, respectively. The analysis was carried out by the aid of the computer package (STATISTICA 12) for Windows. The level of significance used was $p < 0.05$.

3. Results and Discussion

The present study provides new insights into the potential effects of JP at different C:N ratios on tilapia culture water parameters as well as their growth performance, haemato-immune parameters, serum bactericidal activity, and immune-related genes expression profiles, and resistance to bacterial insults. The results of water physical parameters showed considerable variations among the experimental groups throughout the 75 days of the experimental period (Table 2). In all experimental groups, the pH values were varied between 7.3 and 8.2, whereas the range of water temperature was between 27.9 and 28.9 °C. The values were within the allowed limits for tilapia culture, as the standard thermal comfort zone for tilapia ranges from 27 to 32 °C, while the optimal pH values are between 6.0 and 8.5 [40]. The DO values in biofloc treated tanks such as C:N12, C:N15, and C:N20 were varied between 5.8-6.1, 5.5-6.0, and 5.9-6.1 mg/L, respectively. The close reading values between the treatments were due to the implemented

robust ventilation systems used for biofloc maturation. Regular aeration and frequent addition of carbon substrates are necessary to support net primary production, biomass formation and prevent the system collapse [41]. The minimum DO value for tilapia reared in biofloc technology (BFT) is 4 mg/L [42]. Control tanks have significantly higher DO values (6.4-6.6) than the treatment, which is either due to the regular exchange of water or to the absence of bacterial biomass, in agreement with Lima et al. [26].

The results highlighted that the values of TAN, nitrite, and ammonia were considerably reduced, whereas TSS values elevated significantly in all biofloc-treated groups compared to control (Figs. 1 A-D). The TSS values were significantly different ($p < 0.05$) between the treatments and the values were directly proportional to the amount of added jaggery and C: N ratio levels. The highest TSS value was observed in C:N20 (T3) group, followed by C:N15 (T2) and C:N12 (T1), where the relative values at the end of the experiment were 573.70 ± 10.82 , 473.77 ± 11.87 and 402.40 ± 3.78 mg/L, respectively (Fig. 1A). Our results bear a close resemblance to Dauda et al. [43] who reported that the TSS value at C:N20 was significantly higher than C:N10 in week 4 and towards the end of the experiment. TSS levels up to 500 mg/L are convenient for fish culture [44]. Accordingly, the current TSS values were at the optimum level or increased flexibly, thus explained the high levels of dissolved oxygen concentration during the experiment, in agreement with Gaona et al. [45].

Interestingly, the levels of TAN, ammonia, and nitrite were remained constant and close to the permissible limit for fish farming throughout the experiment in all biofloc-treated groups, while the values were increased dramatically over time in control (Fig.1 B-D). The significant reductions in all values were in a linear proportion with the amount of jaggery added and necessarily correlated with the floc volume, an abundance of heterotrophic bacteria, and plankton density (Fig.1 E-G). Several other studies clarified the key role of heterotrophic bacteria and plankton population in denitrification, organic matter decomposition and mineralization [46]. A previous report on marine studies evidenced the ability of heterotrophic bacteria to utilize 42% and 16% of ammonia and nitrate, respectively [47]. The biofloc volume and the level of toxic ammonia depend on; 1) the carbon content of the selected carbohydrates, 2) the C:N ratio, and 3) the microbial conversion coefficient [48]. Avnimelech [49] mentioned that the carbon source supplement could reduce the level of dietary protein and thereby reduce the TAN value. Carbon amendment and C:N ratio in the biofloc system is crucial to regulating inorganic nitrogen and the

eventual growth of bacterial biomass, which refers to the availability and competition for organic carbon and ammonia [50].

In this study, the control group (bioflocs free group) utilized 1.2 m³, equivalent to 0.4 m³ for each tank; however, the treated groups did not receive any water exchange, except for correction of evaporation losses, indicating the potential role of the current system in sustainable biomass production and water conservation. Broadly, biofloc technology (BFT) provides a potential way for the maintenance of the water quality within a favorable scope for most aquatic species [51]. The bioflocs were harvested at the end of the experiment and exhibited various proximate compositions (Table 3). The data indicated that all the nutritional constituents of the bioflocs were significantly influenced by the addition of OJ, as many essential, nonessential, and conditionally essential amino acids were recorded (Table 4). The values were presented as the protein level (%) and found close to the recommended levels [52]. Similar findings were reported by Durigon et al. [53] who regulating the level of digestible protein and energy in diets of Nile tilapia reared in a biofloc technology.

The content of crude protein, crude fat, and raw fiber increased significantly after the treatment in a dose-dependent manner. The highest values were noticed in T3 group (C:N20), where the proximate analysis (based on the dry matter) showed 6.01±0.64% for crud lipid, 51.90±1.85% for crude protein, and 17.10±0.63 for crud fiber. Moreover, Tacon et al. [54] estimated the proximate composition of shrimp reared bioflocs as; a) 5–9% crude lipid, b) 35–38% crude protein, and c) 7–10 % ash. Chou and Shiau [55] reported that 5% of dietary lipids seemed adequate to satisfy the basic needs of juvenile hybrid tilapia (*O. niloticus* × *O. aureus*). For ash and moisture contents, the values were not influenced by the treatment, and not all exceeded the desirable limits for fish farming, in agreement with Azim and Little [29].

In this work, fish exposed to different concentrations of jaggery showed a significant enhancement ($p < 0.05$) in all growth performance parameters compared to the control (Table 5). T1 and T2 groups showed no statistically significant differences between final weight gain (WG), food conversion ratio (FCR), and specific growth rates (SG %), however, they exhibited significantly reduced values ($P < 0.05$) compared to T3 group. Although all the experimental groups received the same basal diet (30% CP), fish raised at different C:N ratios provided higher WG and lower FCR, indicating that the quality of the biofloc does not depend entirely on the basal food. This finding is in good accordance with the findings of Azim and Little [29] who did

not notice any significant difference in all nutritional parameters while using basic diet regimens with different CPs (35% and 24%) in the biofloc system. Fish are grown in bioflocs at different C:N ratio provided comparatively higher average final weight, which ranging from 1.08 ± 0.02 to 1.20 ± 0.02 than that of the control (0.85 ± 0.04). These results are consistent with those observed by Azim and Little [29], while they are in contrast with the results of Lima, Souza, Girao, Braga and Correia [26], where the control fish grew better than those cultivated in the liquid molasses-based biofloc. Overall, significant increases in all growth criteria were attributed to the superior level of crude protein and raw fat contents of biofloc, and their amino acid composition, in consonance with Yun, Shahkar, Hamidoghli, Lee, Won and Bai [56]. The nutritional effect of the biofloc is not entirely dependent upon its content but is largely based on the capability of fish to digest and absorb these suspended particles [22].

In addition to that, BFT also supported for the greater survival rate in tilapia. It was noted that the higher the level of carbon source, the greater the survival rates of fish. Fish reared with high C:N ratio (C:N20) exhibited approximately 3.3% end mortality rate, whereas the fish in the control group reported significantly higher mortality up to 16.67% (Table 5). Thus, it could be related to the high load of toxic ammonia (0.078 ± 0.0078) and nitrite (0.87 ± 0.045) in the control tanks due to absence of biofloc and infrequent water exchange. The high levels of ammonia in water might affect the accumulation of ammonia in fish blood and tissues, and thus negatively affect their growth and survival rate [57]. However, the addition of OJ greatly improves the evolution of the microbial community and other primary organisms such as planktons, rotifers, and diatoms, which utilize ammonia and nitrite in biofloc tanks and provide a natural food source for fish farming [48]. Similar survival kinetic was reported in shrimp biomass systems after using the same carbon substrate [22].

Assessment of hematological indices is a fundamental intrinsic tool for the evaluation of fish welfare [32]. The results of hematological investigations suggested that the total WBCs, RBCs, and HCT values were increased with increasing the concentration of jaggery, and the maximum was observed with the T3 group. Furthermore, an unexpected significant reduction in hemoglobin values was observed in all treated groups compared to the control (Table 6). The clear results of the present study are in good agreement with Xu and Pan [12] who noticed a significant increase in the blood indices of shrimp reared in biofloc system. In contrast to our results, Azim and Little [29] did not score a significant difference in hematocrit values of Nile

tilapia between the control and biofloc-treated groups. A similar trend in RBCs count was reported in the *O. niloticus* and *Litopenaeus vannamei* reared in different biofloc systems [58, 59]. Erythrocytes are indicators of different types of stress and their concentration in the blood is directly proportional to the availability of oxygen within the tissues [60]. Furthermore, Leukocytes (WBCs) are the most involved cells in fish innate immunity and their abundance reflects the fish's health status [33]. Therefore, the relatively low levels of RBCs and WBCs in the control group indicated the stressful effect of higher ammonia and nitrite and explained the possible causes of high mortality in the same group. Several studies related to aquaculture nutrition have utilized a high-quality complex diet not only for optimal growth but also to improve the immunological, antioxidant, and physiological conditions of fish [61]. To present the overall status of the Nile tilapia antioxidant defense system, the level of different antioxidant enzymes was evaluated in the blood. SOD, CAT, and GSH are intrinsic antioxidant and immune-related enzymes designed to address oxidative stress and enhance the fish immunity [62, 63]. The enzymes are mainly involved in lipid peroxidation and act collectively against the generated free radicals to protect the active cells from deterioration and resisting aging [27]. Besides, GSH it is the main enzyme responsible for the balance of cellular redox status [64].

The present study demonstrated that the fish reared under jaggery-based BFT (JB-BFT) exhibited remarkably a higher antioxidants activity than the fish maintained in clear-water. It was revealed that the potential effect of biofloc on the level of antioxidants seemed to be carbon ratio-dependent (Table 7), in agreement with Liu et al. [65] who observed significant increases in superoxide dismutase, glutathione, and malondialdehyde based on the supplemented CN ratios, where the highest values were recorded in the CN 15 and CN 10 groups compared to the control group ($p < .05$). As indicated, the serum SOD activity showed an improvement in all treated groups compared to control, which could be attributed to the biologically active derivatives of mature biofloc [58]. The highest values of SOD enzyme activity were recorded in the T3 group. Similarly, higher absolute values of CAT and GSH were recorded in the same group; however, their kinetics did not change significantly in T1 and T2 groups. CAT and GSH play a vital role against lipid peroxidation, as it converts active free oxygen radicals into non-toxic compounds [64, 65]. A sudden drop in both values promotes free radical accumulation, leading to injury and disruption of cell functions [66, 67].

Herein, the increased level of antioxidant enzymes in the treated fish reflects the efficiency of BFT and also indicates fish wellbeing and low oxidative stress [68]. Our findings are in agreement with Mansour and Esteban [69] who observed a significant increase in SOD and glutathione activities of Nile tilapia being cultured in the BFT system. Oxidative stress in fish is a result of the imbalance between pro-and antioxidant species [70]. Thus, in this study, the increased levels of MDA in the control group reflected the high level of lipid peroxidation and suggested evidence of exhaustion of tilapia antioxidant defense as a result of ammonia and/or nitrite toxicity [71]. The MDA enzyme has adverse toxic effects on cells and could induce tissue damage [72], and thus explained why a relative decrease in survival rate was observed in the control group.

Results of the innate immune parameters evidenced that there was an increase in serum lysozyme, protease, and antiprotease activity after adding various concentrations of jaggery and the increase was noticed predominantly in fish given the high dose of jaggery (Table 7). However, there were no significant variations between T1 and T2 groups, except for antiprotease activity. This lack of variation could be related to the proximity of C:N lineage, which was insufficient to provide a substantial difference in the innate immune functions. Several other researchers were also demonstrated about the immunomodulatory influence of the BFT system [73, 74]. Biofloc contains heterogeneous bacteria, plankton, protozoa, and other molecules that help to activate nonspecific fish immunity through the PAMP recognition system [75]. Moreover, the bacterial community formed in BFT is a good source of many organic and antibacterial compounds, and it also embraces many cellular components and active metabolites, which have a potential effect on fish immunity [76, 77]. In addition, the increased level of serum lysozyme in the present study was coincident with the surge of white blood cells, because neutrophils are the main intrinsic source of this enzyme [78]. Our results are compatible with Verma et al. [74] who observed an improvement in the immune parameters of Rohu fingerlings grown in tapioca-based biofloc. Overall, the considerable diversity in serum lysozyme, protease, and antiprotease activity categorically suggested about the potential impact of jaggery on the tilapia immune response by the favorable activation of some immune cells and enzymes.

The efficacy of the BFT developed in this study was assessed in terms of protection; the serum bactericidal capacity of the treated fish was evaluated against the major aquatic threatening pathogens such as *A. hydrophila* and *S. agalactiae*. The results suggested that the long-term

supplementation of jaggery was significantly enhanced the serum bactericidal activity against the tested pathogens (Table 7). The highest serum bactericidal activity was observed in the T3 group with an average percentage of 31.97 ± 1.61 and 38.71 ± 0.98 for *A. hydrophila* and *S. agalactiae*, respectively. Our results were in line with the earlier findings of Verma et al. [74], who noticed a significant enhancement in the resistance of *L. rohita* grown in the biofloc-based system to *A. hydrophila*. These results gave a concern to the prospective use of JB-BFT as an effective rearing system to combat most bacterial invasions. In terms of production, it is likely that the raised in JB-BFT showed strong serum activity, higher resistance to bacterial insults, and are of great importance in freshwater tilapia culture, as tested bacteria are the main stress-causing pathogens in aquaculture [79, 80]. In the present study, the expression of different immune-related genes (LYS, TNF- α , and IL-1 β) was analyzed in the head kidney of the experimental fish, as the kidney is one of the main endocrine and hematopoietic-lymphoid organs in teleosts [81]. It was noticed that there was a significant increase in relative mRNA expression of target genes ($p < 0.05$) in the biofloc treated fish compared to control, which indicated their influence in immune response activation. TNF- α , and IL-1 β are the main pro-inflammatory cytokines, which play multiple functions including the regulation of cell proliferation, apoptosis and transcriptional regulation [82]. All the encoding genes were upregulated in proportion to the different C:N ratios, with the highest fold increase, observed in the C:N20 group, followed by C:N15 and C:N12 groups (Fig. 2).

Furthermore, the expression of IL-1 β and TNFs genes were significantly increased, which reflects the effectiveness of JB-BFT to modulate the immune response of tilapia and increase its resistance to bacterial infection. IL-1 β stimulates the release of cytokines from activated phagocytes during tissue injury and microbial invasion, while TNFs play a critical role in acute inflammation, lymphoid cell proliferation, and immune cells activation [83]. Our results nearly agree with those of Menaga et al. [84] who observed an increase in the expression levels of cathepsin L, TLR7, and IL-1 β immune genes in the head kidney of tilapia grown in biofloc system. Kheti et al. [85] reported that the nutritional supplement of microbial floc increased the expression ratios of IL-1 β and TNF- α in the head kidney and liver of *Rohu potentiates*. A similar trend of target genes expression was recorded in the intestinal tissue of Nile tilapia after a dietary combination of *Echinacea purpurea* extract and vitamin C [86]. In this study, fish injected with sterile saline (C -ve) showed no mortalities or pathological lesions, while those challenged with

A. hydrophila displayed pathopnomonic clinical signs and postmortem findings on the second day post-infection. Most of the infected fish showed pop eyes, abdominal distention, fin erosion, petechial hemorrhages, and skin discoloration, consistent with the findings of Rey et al. [87] and Algammal et al. [88]. Signs are mainly due to bacterial toxins and extracellular enzymes, which have a deadly and wobbly effect on fish by inducing massive degenerative changes, tissue damage and cell necrosis [89].

The cumulative mortalities were considerably lower in biofloc treated groups than control (Fig. 3), and the maximum relative protection was recorded in the T3 group given a high C:N ratio (Table 8). It was noted that all untreated fish died within 3-5 days post-challenge, while biofloc-treated fish required a long time 7 to 13 days, each independently according to the C: N ratios. Based on the results of the challenge trial and previous findings of serum bactericidal activity, jaggery-based biofloc technology is considered as a potential alternative strategy for controlling pathogens, including *A. hydrophila*. The basic principle of biofloc treatment is to fortify the fish innate immunity and thereby increased their resistance to invading pathogens [90]. The presented results are in a good agreement with the previous findings of Ekasari et al. [10] and Dauda et al. [43].

4. Conclusion

JB-BFT encouraged the ammonia immobilization via biomass maturation and nitrification. The volume of the flocs and the overall nutritional compositions are directly proportions to the amount of jaggery added. The C:N20 biofloc groups were considered as more appropriate for tilapia culturing, as fish raised in this group provided better growth performance, higher survival rate, enhanced antioxidant capacity, serum bactericidal activity against bacterial insults, and higher resistance against *A. hydrophila* infection. Based on these results, JB-BFT has a robust influence on Nile tilapia innate immunity through favorable innovation of various immune-cells and enzymes as well as upregulating the immune genes expression profile. More importantly, the present study offers jaggery as a potential new carbon source with unique and distinct properties that satisfy all considerations of biofloc technology with high productivity and zero water exchange.

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Table 1. List of primers used for immune-genes expression.

Gene name	Acronym	Function	NCBI Genebank (Gene ID)	Sequence (5'-3')
Beta-actin	β -actin	Housekeeping gene	100534414	F:CCACACAGTGCCCATCTACGA R:CCACGCTCTGTCAGGATCTTCA
lysozyme C	LYZ	Immune-related genes	102075604	F: GGAGGCAGTGAAGACCCAAA R: TCCCACGCACAAATGGAAGT
Tumour necrosis factor alpha	TNF- α		100692128	F: GCTACGACTCCCAGCACTTTG R: GCGGTACTGCTCGGATCTCT
Interleukin 1 beta	IL-1 β		100707066	F: TGTCGCTCTGGGCATCAA R: GGCTTGTCGTCATCCTTGTGA

Table 2. Average values of water temperature (T °C), dissolved oxygen (DO), and pH of control and BFT-based nursery-reared Nile tilapia (*O. niloticus*) throughout the experiment (75 days).

Parameter	Units	C [†]	T1 [‡]	T2 [§]	T3 ^π
Temperature	(° C)	28.3-28.9	28-28.6	27.9-28.5	27.9-28.6
DO	(ppm)	6.4-6.6	5.8-6.1	5.5-6.0	5.9-6.1
pH	-----	7.8-8.2	7.3-7.6	7.3-7.6	7.4-7.8

[†]C, Nile tilapia (*O. niloticus*) reared in water culture system without carbon substrates. [‡]T1, [§]T2, and ^πT3, Nile tilapia (*O. niloticus*) reared in a biofloc system using jaggery as a carbon substrate at different C:N ratios of 12:1, 15:1, and 20:1 respectively.

Table 3. Proximate composition of control basal diet and biofloc obtained from BFT-based nursery-reared Nile tilapia (*O. niloticus*) at 75 days of the experimental period.

% Dry mater	C [†] (feed)	T1 [‡]	T2 [§]	T3 ^π
Crude lipid	3.13±0.25 ^b	6.25 ± 0.59 ^a	6.67 ± 0.81 ^a	6.01 ± 0.64 ^a
Crude protein	29.43±0.81 ^d	38.50± 1.18 ^c	42.97± 0.31 ^b	51.90±1.85 ^a
Crude fiber	12.19±0.41 ^c	15.63±0.69 ^b	16.01±0.27 ^{ab}	17.10±0.63 ^a
Ash	15.08±0.31 ^b	10.67±0.90 ^a	11.61±1.00 ^a	11.73±0.73 ^a
Moisture	7.90±0.82 ^a	5.12±0.18 ^{bc}	5.91±0.11 ^b	4.31±0.46 ^c

Data are presented as means ± SD ($n = 3$). Superscripted letters denote significant differences between experimental groups (ANOVA; $p < 0.05$). [†]C, Nile tilapia (*O. niloticus*) reared in water culture system without carbon substrates. [‡]T1, [§]T2, and ^πT3, Nile tilapia (*O. niloticus*) reared in a biofloc system using jaggery as a carbon substrate at different C:N ratios of 12:1, 15:1, and 20:1 respectively.

Table 4. Amino acid composition (protein level %) of biofloc obtained from BFT-based nursery reared Nile tilapia (*O. niloticus*) at 75 days of the experimental period.

Amino acids (AA)	Protein level (%)		
	T1 [‡]	T2 [§]	T3 ^π
Essential AA			
Histidine	1.45±0.03	1.49±0.03	1.49±0.04
Arginine	3.73±0.07 ^c	3.91±0.06 ^b	4.10±0.06 ^a
Isoleucine	4.14±0.07 ^b	4.23±0.10 ^{ab}	4.36±0.08 ^a
Leucine	6.44±0.65	7.08±0.92	6.15±0.07
Lysine	4.32±0.08	4.39±0.08	4.46±0.13
Methionine	1.34±0.06	1.33±0.06	1.39±0.08
Cysteine	1.82±0.08	1.86±0.08	1.90±0.07
Valine	5.08±0.51	4.46±0.07	5.18±0.52
Phenylalanine	5.01±0.10 ^a	5.07±0.10 ^a	4.11±0.08 ^b
Threonine	4.99±0.13	5.01±0.13	5.05±0.09
Tryptophan	0.57±0.09	0.59±0.08	0.62±0.09
Non-essential AA			
Aspartic Acid	10.27±0.08	10.35±0.11	10.50±0.16
Glycine	8.05±0.13	8.10±0.12	8.14±0.09
Glutamic Acid	11.63±0.51	11.35±0.15	11.44±0.21
Alanine	6.79±0.19	6.89±0.10	7.00±0.19
Hydroxylysine	0.45±0.13	0.47±0.06	0.48±0.06
Serine	3.66±0.38	3.64±0.38	3.83±0.37
Tyrosine	2.99±0.11 ^b	3.10±0.02 ^{ab}	3.39±0.25 ^a
Ornithine	0.59±0.03 ^b	0.64±0.03 ^b	0.78±0.02 ^a
Conditionally essential AA			
Proline	4.06±0.05 ^c	5.00±0.13 ^b	5.42±0.23 ^a
Taurine	0.22±0.02 ^b	0.29±0.02 ^a	0.35±0.03 ^a
Cysteine	0.96±0.02 ^c	1.20±0.06 ^b	1.98±0.11 ^a
Phenylalanine	4.28±0.61 ^b	5.03±0.16 ^{ab}	5.04±0.41 ^a
Hydroxyprolin	0.56±0.05 ^b	0.66±0.06 ^b	0.79±0.03 ^a

Data are presented as means ± SD ($n = 3$). Superscripted letters denote significant differences between experimental groups (ANOVA; $p < 0.05$). [‡]T1, [§]T2, and ^πT3, Nile tilapia (*O. niloticus*) reared in a biofloc system using jaggery as a carbon substrate at different C:N ratios of 12:1, 15:1, and 20:1 respectively.

Table 5. Growth performance of Nile tilapia (*O. niloticus*) reared in control and BFT-based culture systems at 75 days of the experimental period.

Parameters	Units	C [†]	T1 [‡]	T2 [§]	T3 ^π
Initial weight	(g/biomass)	515.31±3.90	506.33±5.13	506.29±5.42	504.23±5.53
Body weight gain	(kg/biomass)	0.85±0.04 ^c	1.08±0.02 ^b	1.09±0.01 ^b	1.20±0.02 ^a
Feed consumed	(kg/biomass)	1.99±0.04 ^a	1.54±0.13 ^b	1.52±0.13 ^{bc}	1.27±0.06 ^c
Feed conversion ratio	FCR	2.36±0.11 ^a	1.43±0.10 ^b	1.39±0.11 ^b	1.06±0.04 ^c
Specific growth rate	SGR	1.29±0.04 ^c	1.52±0.02 ^b	1.53±0.01 ^b	1.63±0.03 ^a
Survival rate	%	83.33±2.89 ^b	90±0 ^{ab}	93.33±5.77 ^a	96.67±2.89 ^a

Data are presented as means ± SD ($n = 3$). Superscripted letters denote significant differences between experimental groups for a given time (ANOVA; $p < 0.05$). [†]C, Nile tilapia (*O. niloticus*) reared in water system without carbon substrates. [‡]T1, [§]T2, and ^πT3, Nile tilapia (*O. niloticus*) reared in a biofloc system using jaggery as a carbon substrate at different C:N ratios of 12:1, 15:1, and 20:1 respectively.

Table 6. Hematological profiles of Nile tilapia (*O. niloticus*) reared in control and BFT-based culture systems at 75 days of the experimental period.

Parameters	Units	C [†]	T1 [‡]	T2 [§]	T3 ^π
WBCs [¶]	(10 ⁴ /μl)	5.53±0.57 ^c	6.61±0.25 ^b	7.06±0.07 ^{ab}	7.63±0.44 ^a
RBCs ^{¶¶}	(10 ⁶ /μl)	1.82±0.17 ^b	2.15±0.06 ^b	2.01± 0.03 ^b	2.62±0.27 ^a
Haemoglobin	(g/dl)	5.48±0.28 ^b	7.64±0.20 ^a	7.99±1.34 ^a	8.10±1.58 ^a
HCT ^{¶¶¶}	(%)	22.10±1.06 ^b	21.67±1.17 ^b	22.43±0.47 ^b	25.27±0.35 ^a

[¶]WBCs, white blood cells; ^{¶¶}RBCs, Red blood cells; ^{¶¶¶}HCT, Hematocrit. Data are presented as means ± SD (*n* = 3). Superscripted letters denote significant differences between experimental groups for a given time (ANOVA; *p* < 0.05). [†]C, Nile tilapia (*O. niloticus*) reared in water culture system without carbon substrates. [‡]T1, [§]T2, and ^πT3, Nile tilapia (*O. niloticus*) reared in a biofloc system using jaggery as a carbon substrate at different C:N ratios of 12:1, 15:1, and 20:1 respectively.

Table 7. Serum innate immune parameters and antioxidant enzymes of Nile tilapia (*O. niloticus*) reared in control and BFT-based culture systems at 75 days of the experimental period.

	Parameters	Units	C [†]	T1 [‡]	T2 [§]	T3 ^π
Antioxidant enzymes	SOD [¶]	(U/mL)	35.84±3.37 ^c	44.14±4.00 ^b	42.96±2.16 ^{bc}	55.90±2.46 ^a
	MDA ^{¶¶}	(nmol/ mL)	3.29±0.08 ^a	1.99±0.20 ^b	2.04±0.07 ^b	1.60±0.13 ^c
	CAT ^{¶¶¶}	(U/mL)	24.27±1.51 ^c	35.30±4.85 ^b	38.11±4.01 ^{ab}	44.19±1.72 ^a
	GSH ^{¶¶¶¶}	(nmol/ mL)	20.78±2.13 ^c	27.95±2.13 ^b	29.29±1.06 ^b	38.24±2.31 ^a
Innate immune parameters	Lysozyme activity	(µg/ml)	6.40±0.45 ^c	7.58±0.57 ^b	8.16±0.16 ^b	9.73±0.41 ^a
	Protease activity	(%)	13.43±0.81 ^c	17.40±0.57 ^b	18.10±0.50 ^b	21.44±0.63 ^a
	Antiprotease activity	(%)	26.82±0.34 ^d	31.72±0.68 ^c	36.07±2.37 ^b	41.68±1.40 ^a
Serum bactericidal activity	<i>Aeromonas hydrophila</i>	(%)	12.37±0.65 ^c	25.53±2.45 ^b	27.20±1.11 ^b	31.97±1.61 ^a
	<i>Streptococcus agalactiae</i>	(%)	16.46±0.58 ^c	23.79±1.26 ^c	29.52±1.12 ^b	38.71±0.98 ^a

[¶]SOD, Superoxide dismutase; ^{¶¶}MDA, Malondialdehyde; ^{¶¶¶}CAT, Catalase; ^{¶¶¶¶}GSH, Glutathione. Data are presented as means ± SD ($n = 3$). Superscripted letters denote significant differences between experimental groups for a given time (ANOVA; $p < 0.05$). [†]C, Nile tilapia (*O. niloticus*) reared in water culture system without carbon substrates. [‡]T1, [§]T2, and ^πT3, Nile tilapia (*O. niloticus*) reared in a biofloc system using jaggery as a carbon substrate at different C:N ratios of 12:1, 15:1, and 20:1 respectively.

Table 8. Results of challenge trial on mortality percentage and RLP percentage following treatment with jaggery-based biofloc technology.

Treatments	Subgroups	C:N ratios	No of infected fish	No of dead fish	Mortality %	RLP%
[†] C	C -ve [¶]	0	10	0	0	-
	I&II [*]		20	16	80	0
[‡] T1	C -ve [¶]	12:1	10	0	0	-
	I&II [*]		20	13	65	18.75
[§] T2	C -ve [¶]	15:1	10	0	0	-
	I&II [*]		20	11	55	31.25
^π T3	C -ve [¶]	20:1	10	0	0	-
	I&II [*]		20	5	25	68.75

[†]C, Nile tilapia (*O. niloticus*) reared in water culture system without carbon substrates. [‡]T1, [§]T2, and ^πT3, Nile tilapia (*O. niloticus*) reared in a biofloc system using jaggery as a carbon substrate at different C:N ratios of 12:1, 15:1, and 20:1 respectively.

[¶]C -ve, control negative subgroups injected with the sterile saline.

^{*}I&II, infected subgroups with *A. hydrophila* at a final concentration of 3×10^8 CFU/ mL.

Figure captions

Fig. 1. The levels of water total suspended solids (TSS, mg/L) (A), total ammonia nitrogen (TAN, mg/L) (B), unionized ammonia (NH₃-N, mg/L) (C), nitrite (NO₂-N, mg/L) (D), heterotrophic bacteria (CFU/mL) (E), floc volume/L (F), and plankton (cells/L) (G) of control and BFT-based nursery reared Nile tilapia (*O. niloticus*) throughout the experimental period (75 days). Data are presented as means $\pm$ SD ($n = 3$). Superscripted letters denote significant differences between experimental groups for a given time (ANOVA; $p < 0.05$). C, Nile tilapia (*O. niloticus*) reared in water culture system without carbon substrates. T1, T2, and T3, Nile tilapia (*O. niloticus*) reared in a biofloc system using jaggery as a carbon substrate at different C:N ratios of 12:1, 15:1, and 20:1 respectively.

Fig. 2. Immune-related gene expression profiles in head kidney of Nile tilapia (*O. niloticus*) reared in control and BFT-based culture systems at 75 days of experimental period. Data are presented as means $\pm$ SD ($n = 3$). Asterisk denotes significant differences between experimental groups (ANOVA; $p < 0.05$). T1, T2, and T3, Nile tilapia (*O. niloticus*) reared in a biofloc system using jaggery as a carbon substrate at different C:N ratios of 12:1, 15:1, and 20:1 respectively.

Fig. 3. Cumulative mortality (%) over time (Days 0–14) for *Oreochromis niloticus* reared in clear water system (C) and jaggery-based biofloc systems at different ratios of C:N12, C:N15 and C:N20 (T1, T2 and T3, respectively) after experimental challenge with *A. hydrophila*;

Fig. 1 A-D

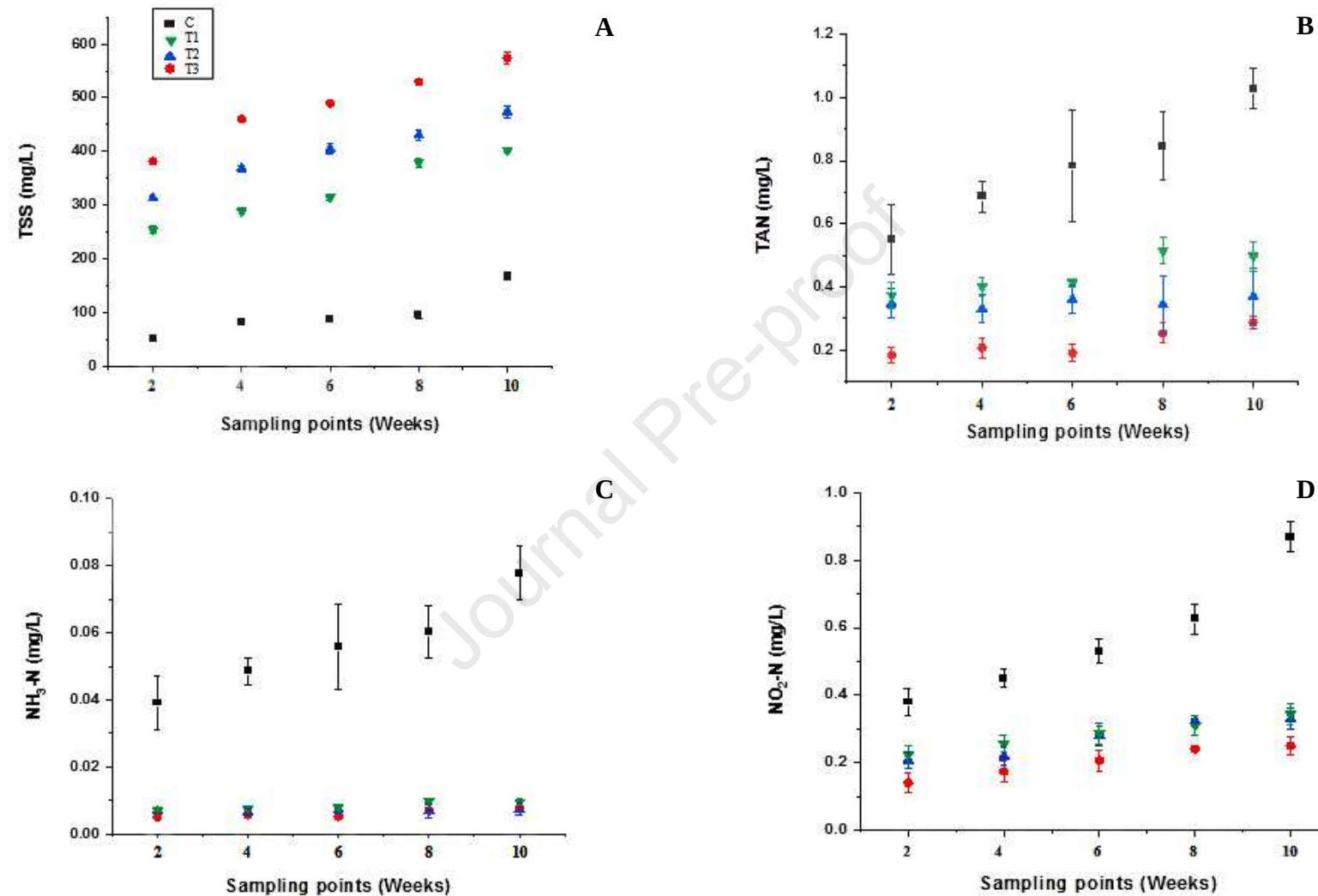


Fig. 1 E-G

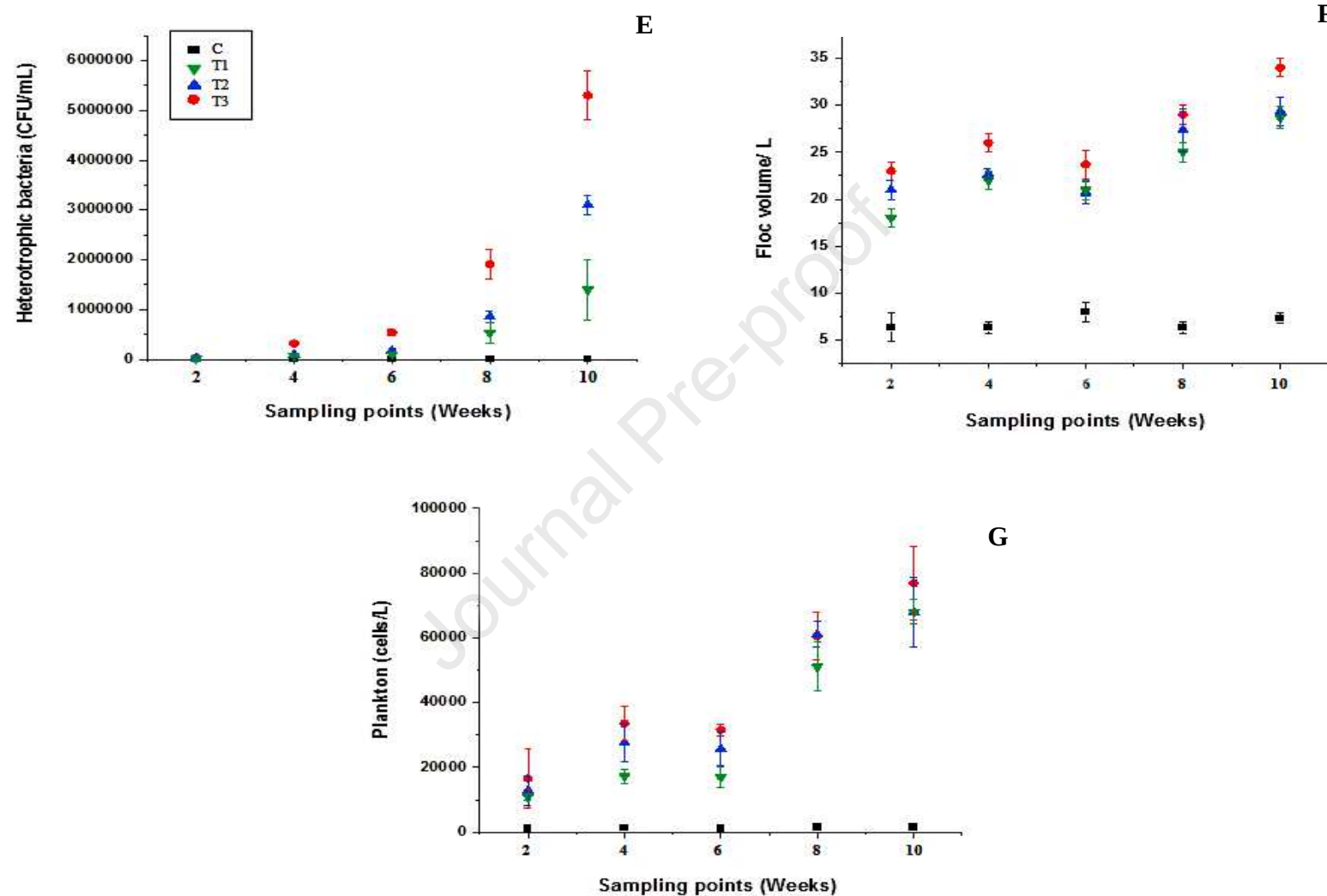


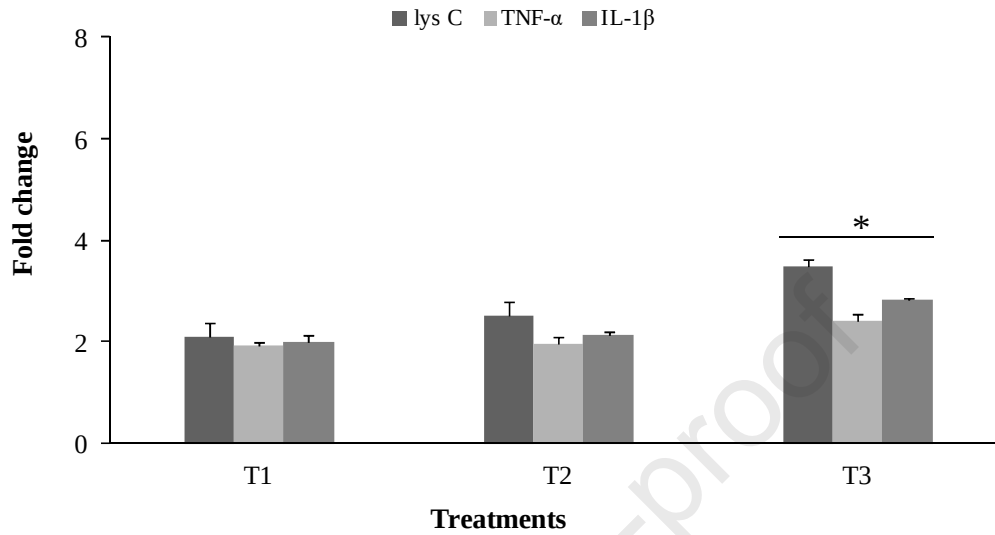
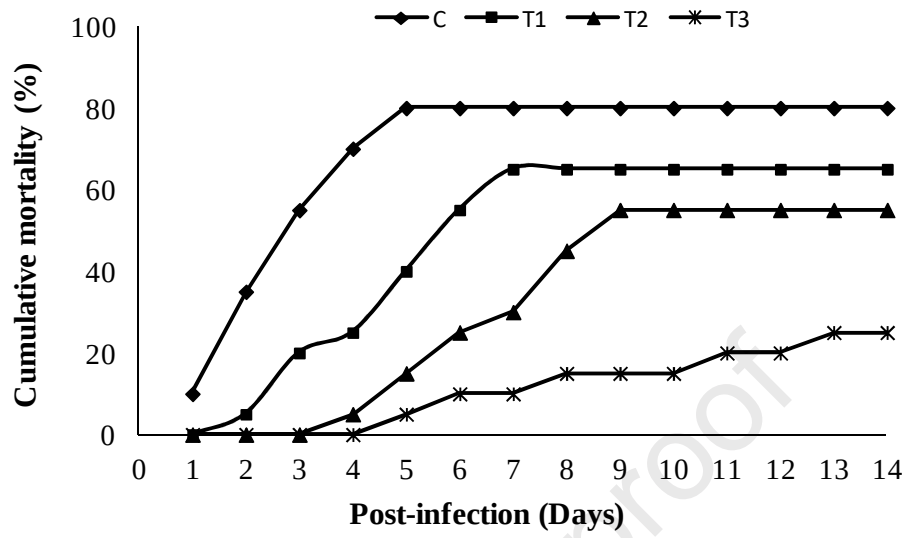
Fig. 2.

Fig. 3.

Highlights

- Jaggery is a new potential carbon source with unique and distinct properties.
- Jaggery-based biofloc technology (JB-BFT) encourages ammonia immobilization and improves the overall water quality through bacterial assimilation and nitrification
- C: N20 provides better growth performance, higher survival rate, enhanced antioxidant capacity, and higher resistance to *Aeromonas hydrophila* infection.
- JB-BFT has a robust influence on Nile tilapia innate immunity through positive innovation of various immune-cells and enzymes as well as upregulating the immune genes expression.
- JB-BFT satisfies all considerations of biofloc technology in an eco-friendly manner.